NEXT AUCKLAND SCOOTER PROJECT

Feasibility & Execution Strategy Report for Timely Market Launch

Prepared for:
Ben Keith *– Product Manager*
Next Auckland Company (NAC)

Prepared by:
Thi Nga Doan
BUSINFO710 Green Consultant

Executive Summary

The **Next Auckland Scooter Project** aims to launch a new children's scooter by 6 October 2025, targeting the peak Christmas sales season. From the start date (3 March 2025) to the deadline, there are 150 working days available.

At first glance, the project seemed feasible. However, detailed analysis revealed resource constraints—particularly the limited availability of Industrial Engineers and the single Purchasing Agent—that prevent key tasks from running in parallel. This forces sequential execution, extending the timeline to 153 working days, missing the deadline by 2 days.

By revising task sequencing and allocating additional resources only to critical-path tasks the project duration improves to 128 working days, equivalent to about 25.5 weeks. The project would finish on 3 September 2025, creating a 1-month safety buffer before the required launch date of 6 October 2025.

Further findings show:

- 11 out of 13 tasks are on the critical path, making the schedule highly sensitive to any delay.
- Cash flow analysis revealed a mid-June cost spike, which can be smoothed by delaying non-critical Task 6 (Detailed Marketing Plan) by 40 days without affecting the finish date.

Recommendations:

- Adopt the optimized plan (Scenario 2), allocating additional resources only to critical-path tasks to secure a one-month safety buffer.
- Monitor all critical-path activities closely to prevent delays that could directly affect the final delivery.
- Support and treat staff with care, providing necessary training or guidance to maintain productivity and quality standards.
- Leverage the slack in Task 6 to smooth cash flow and balance workload without impacting the overall schedule.
- Communicate milestones clearly with stakeholders and conduct regular plan reviews to quickly address any risks or changes.
- Invite contracted resources to the Celebrate day to acknowledge their valuable contribution, strengthening relationships for future collaboration

This optimized approach aligns with NAC's Time Enhanced, Scope Constrained, Cost Acceptable priority matrix, ensuring a timely and quality-assured launch before the 2025 Christmas sales window.

Contents

1. Introduction

The **Next Auckland Scooter Project** is a new product introduction project aimed at launching a children's scooter by 6 October 2025, strategically timed to capture the peak Christmas sales window. In the highly competitive and volatile toy market, missing this seasonal demand would significantly reduce sales potential and market impact.

The project's Priority Matrix defines Time as Enhanced, Scope as Constrained, and Cost as Acceptable—all required tasks must be completed without reducing scope, while additional spending is justified if it helps meet the launch deadline.

The project consists of 13 activities, structured into three major phases:

- Phase 1: Market & Design (Tasks 2–5) – Market Analysis, Product Design, Manufacturing Study, and Product Design Selection.
- Phase 2: Development & Prototyping (Tasks 6–9) – Detailed Marketing Plan, Manufacturing Process, Detailed Product Design, and Test Prototype.
- Phase 3: Finalization & Production Setup (Tasks 10–13) – Finalized Product Design, Ordering Components & Equipment, and Installation of Production Equipment, followed by the final Celebrate milestone (Task 14)

From 3 March 2025 to 6 October 2025, the project spans 218 calendar days, which translates to 150 working days (Monday–Friday, excluding New Zealand public holidays). Optimizing the project duration not only ensures the deadline is met but also creates valuable buffer time for trial production, quality assurance, and launch event preparation, reducing last-minute risks.

This report evaluates the feasibility of delivering the project within the available timeframe, identifies key resource conflicts, and proposes optimized scheduling and resource strategies. It also analyzes alternative execution scenarios, cost implications, and cash flow distribution, providing strategic recommendations to ensure an on-time, quality-assured launch.

2. Problem and Objectives

2.1 Problem

The project faces resource constraints that create scheduling conflicts and extend the timeline beyond the planned launch date. Limited availability of Industrial Engineers, Marketing Specialists, and a single Purchasing Agent prevents several tasks from running in parallel, risking a missed market opportunity and reduced sales potential for the 2025 Christmas season.

2.2 Objectives

Determine the feasibility of meeting the 6 October 2025 launch date within the available timeframe.

Resolve resource overallocations through revised task sequencing and, where necessary, strategic use of additional resources.

Minimize project duration while maintaining full scope and ensuring quality.

Assess cost impacts and cash flow distribution to ensure financial manageability.

Provide clear, actionable recommendations to optimize schedule, cost, and risk management in line with NAC's Time Enhanced, Scope Constrained, Cost Acceptable priority matrix.

3. Analysis of the Business Scenario

3.1 Resource Constraints and Leveling Analysis

The initial plan proposed by Ben appeared feasible when considering only task durations and predecessor relationships. The total planned completion time was 116 working days, well within the 150 working days available from 3 March to 6 October 2025. The schedule accounted for a Monday–Friday workweek and observed New Zealand public holidays (Good Friday, Easter Monday, Anzac Day, King's Birthday, and Matariki).

However, once human resource availability was factored in, several conflicts emerged due to limited staffing capacity, preventing planned parallel execution. Resource leveling is therefore required to align task schedules with actual resource limits.

Key resource conflict groups identified during leveling:

1. **Product Design vs. Manufacturing Study (Tasks 3 & 4)**
   - Both tasks were originally scheduled to start simultaneously.
   - Combined demand = 5 Industrial Engineers, but only 4 are available, making true parallel execution impossible.
2. **Detailed Marketing Plan, Manufacturing Process, and Detailed Product Design (Tasks 6, 7 & 8)**
   - Tasks 6 & 8 conflict due to Marketing Specialist demand exceeding capacity.
   - Tasks 7 & 8 conflict due to Industrial Engineer demand.
   - Tasks 6 & 7 have no conflict and can run together.
   - These overlaps force staggered starts rather than concurrent execution, extending the schedule.
3. **Ordering Components vs. Ordering Production Equipment (Tasks 11 & 12)**
   - Both require the same Purchasing Agent, preventing simultaneous ordering.
   - One order must finish before the other begins, creating an additional dependency.

These resource constraints prevent planned parallel activities, causing delays beyond the initial theoretical 116-day schedule. Through resource leveling, some tasks must be rescheduled to run sequentially, increasing the overall project duration.

To evaluate the project's feasibility after leveling, multiple scenarios were analyzed by adjusting task predecessors and strategically leveraging additional resources. This helped identify how different scheduling and resource strategies affect project duration, cost, and the ability to meet the critical launch deadline.

Since the project's Priority Matrix enhances time, the primary objective was to minimize total duration while still respecting resource constraints and maintaining scope.

1. Product Design (Task 3) & Manufacturing Study (Task 4)

- *Original expectation*: Development Engineers preferred both tasks to finish simultaneously for a same-day outcome review.
- *Constraint*: Combined demand = 5 Industrial Engineers, but only 4 available, making parallel execution impossible.
- *Decision*: Run Product Design first, then Manufacturing Study sequentially. Outcome review can still occur before Task 5 (Product Design Selection) without creating resource conflicts.

2. Detailed Marketing Plan (Task 6), Manufacturing Process (Task 7), and Detailed Product Design (Task 8)

- *Conflict analysis*:
    - Task 6 & 8 conflict due to Marketing Specialist demand.
    - Task 7 & 8 conflict due to Industrial Engineer demand.
    - Task 6 & 7 have no conflict and can run together.
- *Dependency consideration*:
    - Task 8 must be finished before Task 9 (Test Prototype) can start.
    - Task 7 cannot proceed to its next stage until Task 9 is complete.
- *Decision*: Run Task 8 first to unlock Task 9 earlier. Then run Task 7 & Task 6 in parallel with Task 9, reducing idle time.

3. Ordering Components (Task 11) & Ordering Production Equipment (Task 12)

- *Duration breakdown*:
    - Task 11 = 7 days ordering + 10 days shipping = 17 days total.
    - Task 12 = 10 days ordering + 15 days shipping = 25 days total.

- *Scenario comparison*:
  - Doing Task 11 → Task 12 = 32 days total.
  - Doing Task 12 → Task 11 = 27 days total, saving 5 days.
  - Parallel ordering still takes 25 days due to the longer shipping time for equipment.
- *Decision*: Prioritize Task 12 (production equipment) first, then Task 11 (components) to maximize shipping wait time and minimize downstream idle periods.

To better illustrate these changes, Figure 1 and Figure 2 show the task dependency network before and after optimization:

(Detail Network Diagram – see Appendix A)

ID	Task Name
2	Market analysis
3	Product design
4	Manufacturing study
5	Product design selection
6	Detailed marketing plan
7	Manufacturing process
8	Detailed product design
9	Test prototype
10	Finalized product design
11	Order components
12	Order production equipment
13	Install production equipment
14	Celebrate

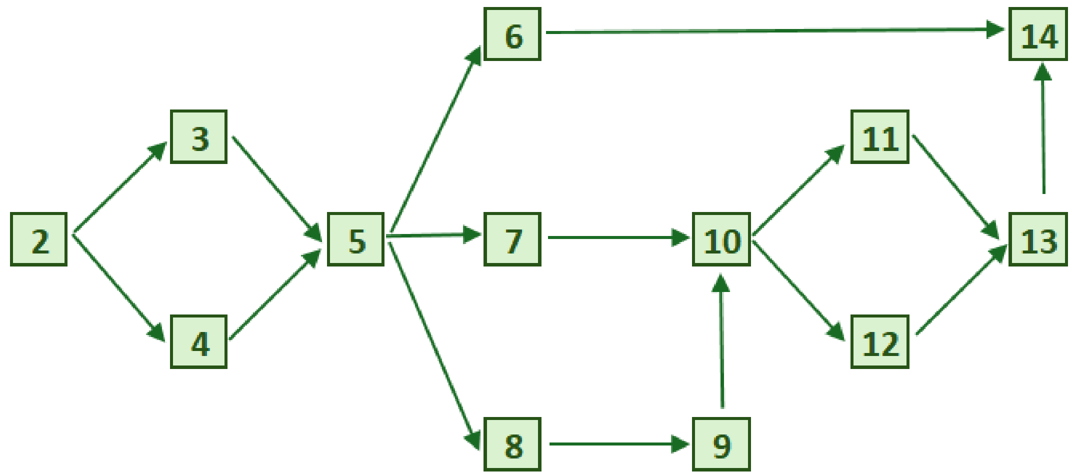


Figure 1. *Preliminary Dependencies (Before Optimization)*

Note:

Original task dependencies (unchanged)

Adjusted dependencies after resource optimization

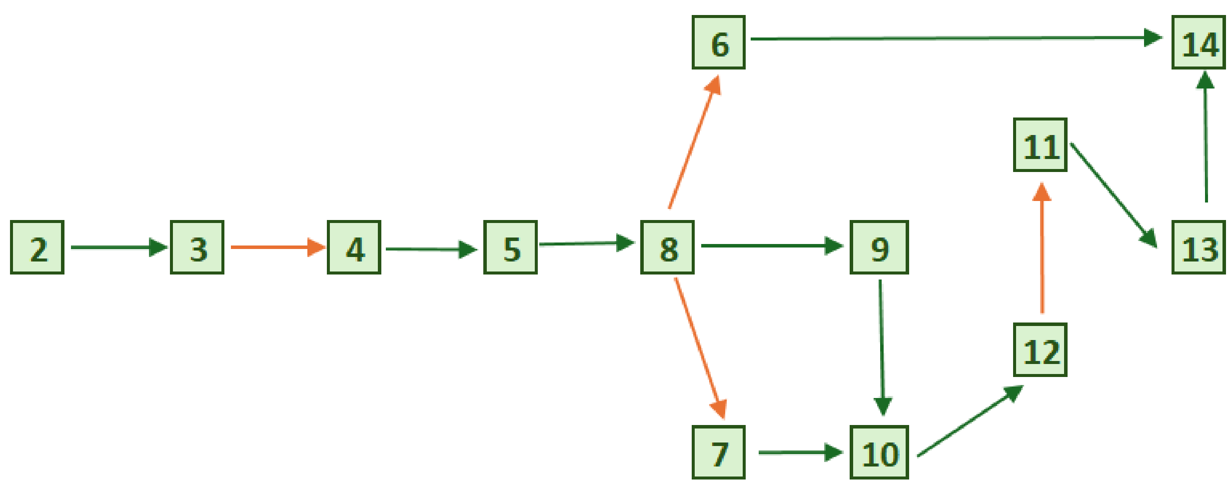


Figure 2. *Resource-Leveled Dependencies (After Optimization)*

3.2 Schedule Feasibility

After revising task predecessors to resolve resource overallocations, the project schedule was re-evaluated using only the original resource availability.

Even with optimized sequencing, several tasks were still forced to run sequentially, extending the timeline beyond the launch date:

- Duration: 153 working days
- Finish date: 8 October 2025
- Outcome: Misses the required launch date by 2 working days.

Because the Priority Matrix enhances time, additional resources were required to further shorten the duration and fully resolve critical overallocations. Four tasks allow additional resource support:

- Detailed Marketing Plan (Task 6)
- Manufacturing Process (Task 7)
- Detailed Product Design (Task 8)
- Install Production Equipment (Task 12)

After applying additional resources—but only to critical-path tasks (7, 8, 12)—the revised schedule improves significantly:

- Duration: 128 working days (approximately 25.5 weeks)
- Finish date: 3 September 2025
- Outcome: Meets the launch deadline with a 1-month safety buffer.

This approach fully resolves overallocated resources on the critical path, enabling a realistic and achievable schedule while preserving the constrained project scope.

3.3 Critical Path

After resource leveling, the project schedule was analyzed to determine its critical path (*see Appendix B for the detailed Gantt chart with start/end dates and visual critical-path highlights*).

Out of 13 total tasks, 11 tasks are on the critical path, meaning any delay will directly impact the final completion date:

- 2: Market Analysis
- 3: Product Design

- 4: Manufacturing Study
- 5: Product Design Selection
- 7: Manufacturing Process
- 8: Detailed Product Design
- 10: Finalized Product Design
- 11: Order Components
- 12: Order Production Equipment
- 13: Install Production Equipment
- 14: Celebrate

Only Task 6 (Detailed Marketing Plan) and Task 9 (Test Prototype) have float (slack). Delays in these two tasks do not affect the overall project duration.

For this reason, additional resources were not applied to Task 6, as accelerating it would not shorten the total schedule. Instead, additional resources were focused only on critical-path tasks (7, 8, 12) where reducing duration directly impacts the finish date.

3.4 Scenario Comparison

Alongside schedule optimization, the cost implications of each scenario were also considered. While the project prioritizes time (enhanced) and scope (constrained), cost remains acceptable, meaning additional spending is justified if it shortens the schedule.

The table below compares the three scenarios, showing their duration, finish date, and total cost:

Scenario	***Duration*** *(working days)*	***Finish Date***	***Total Cost***	***Feasibility***
1 – No additional resources	153	8 Oct 25	$663,920	Misses' deadline
2 – Critical-path tasks only	**128**	**3 Sep 25**	**$595,320**	Meets deadline with 1-month buffer Task 6 slack 52 days, Task 9 slack 12 days
3 – All eligible tasks	128	3 Sep 25	$585,960	Slightly cheaper but no extra time benefit Task 6 slack 60 days, Task 9 slack 12 days

Scenario 3 is only 1.6% cheaper than Scenario 2 but provides no time benefit. From a management perspective, accelerating non-critical tasks with large slack, like Task 6, adds unnecessary effort and coordination without improving the overall schedule.

Focusing only on critical-path tasks (Scenario 2) directs resources where they have the highest impact, achieving the Time Enhanced priority while avoiding unnecessary complexity and cost.

Therefore, Scenario 2 is the most efficient and strategic choice, and it will be used as the basis for all further analysis, including cash flow and workforce utilization.

3.5 Cash Flow Analysis

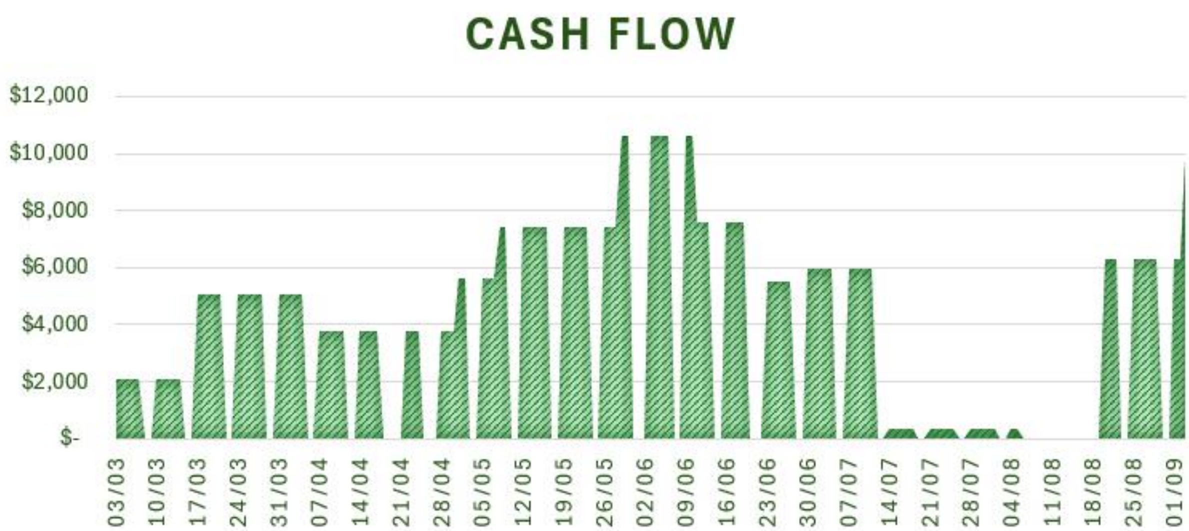


Figure 3: *Peak cost occurs mid-June due to overlapping tasks*

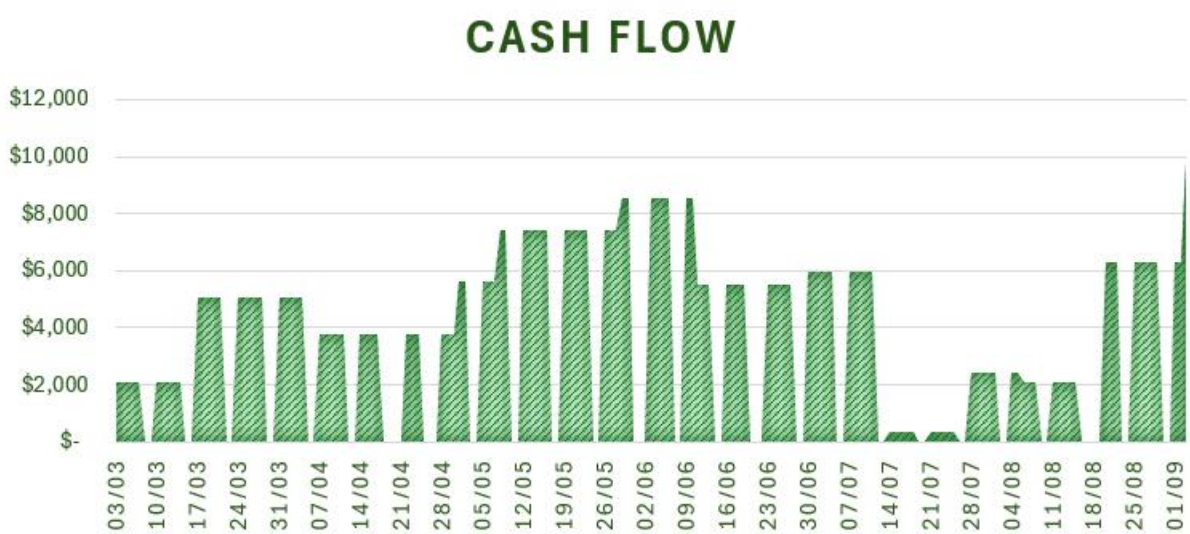


Figure 4: *Delaying Task 6 (40 days) smooths cash flow without affecting the project finish date*

The initial cash flow (Figure 3) shows a pronounced spending peak around mid-June, primarily driven by Task 7 (Manufacturing Process) and Task 8 (Detailed Product Design), each accounting for approximately 20% of total project costs *(see Appendix C).* These high-cost activities overlap with Task 6 (Detailed Marketing Plan) and Task 9 (Test Prototype), concentrating resource demand and creating a temporary financial spike.

By delaying Task 6 by 40 days, which has sufficient slack and does not affect the project's critical path, the workload and expenses are more evenly distributed in the revised cash flow (Figure 4). As a result, the cost peak shifts toward late July, smoothing out financial pressure during the mid-June period.

This adjustment offers two key management benefits:

- Smoother cash outflow, reducing temporary financing pressure during peak periods caused by the high-cost Manufacturing Process.
- Better workforce utilization, avoiding excessive simultaneous task demands while still maintaining the same project finish date.

Since Task 6 is non-critical, delaying it strategically optimizes cash flow timing without risking the overall schedule.

4. Recommendations

Based on the feasibility analysis, the following actions are recommended to ensure the Next Auckland Scooter Project meets its critical launch date and aligns with NAC's Time Enhanced, Scope Constrained, Cost Acceptable priorities:

1. **Adopt Scenario 2 as the execution strategy**
    - Focus additional resources only on critical-path tasks (7, 8, 12) to resolve key overallocations.
    - This approach secures a 1-month safety buffer before the launch date while keeping costs balanced and avoiding unnecessary complexity.
2. **Monitor high-risk tasks closely**
    - With 11 of 13 activities on the critical path, any delay directly affects the project finish date.
    - Pay close attention to these critical activities, as they are the major schedule bottlenecks and pose the highest risk to meeting the launch deadline
3. **Leverage slack to optimize resource and cost timing**
    - Use the available float in Task 6 (Detailed Marketing Plan) to smooth cash flow peaks and reduce simultaneous workload, without affecting the overall finish date.
4. **Plan contingency for resource risks**
    - The bottleneck in Tasks 3 & 4 arises from a shortage of just one Industrial Engineer.
        - If an additional resource can be secured, these two tasks could run in parallel, meeting the Development Engineers' preference for simultaneous completion and saving 15 days overall.
        - If no extra resource is available, allow 1–2 days for outcome cross-checks after each task to ensure quality, aligning with the Scope Constrained priority.
5. **Communicate milestones clearly**
    - Share a high-level phased timeline with stakeholders for clarity, while keeping the detailed Gantt chart as a technical reference for the project team.
    - Invite the additional contracted resources to the Celebrate milestone to acknowledge their contribution, as cost is acceptable under the priority matrix

5. Conclusion

The **Next Auckland Scooter Project** can confidently meet the **6 October 2025 launch date** under **Scenario 2**, finishing one month ahead of schedule. While the schedule is sensitive with most tasks on the critical path, risks can be managed through **focused monitoring of key bottlenecks**, **using slack to smooth cash flow**, and **clear milestone communication**.

This approach aligns with NAC's **Time Enhanced, Scope Constrained, Cost Acceptable** priorities, ensuring a timely and quality-assured launch for the 2025 Christmas sales window.

6. Appendix

6.1 Appendix A – Network Diagram

Figure A1: Preliminary Dependencies (Before Optimization)

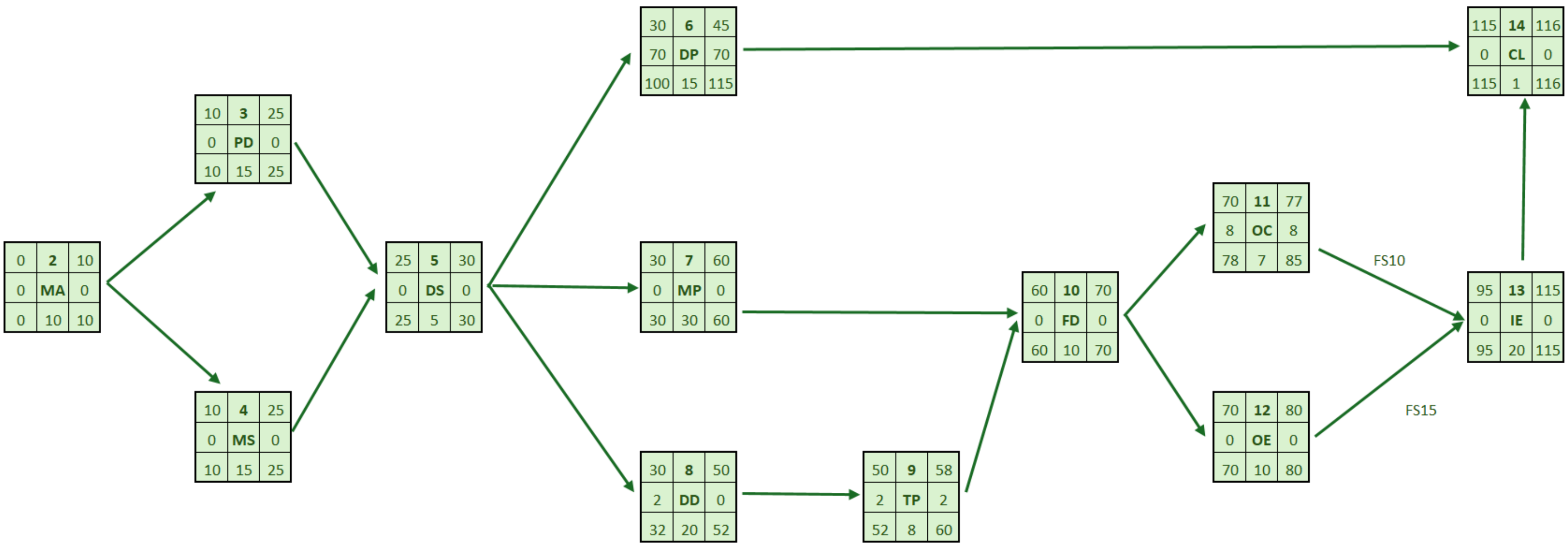
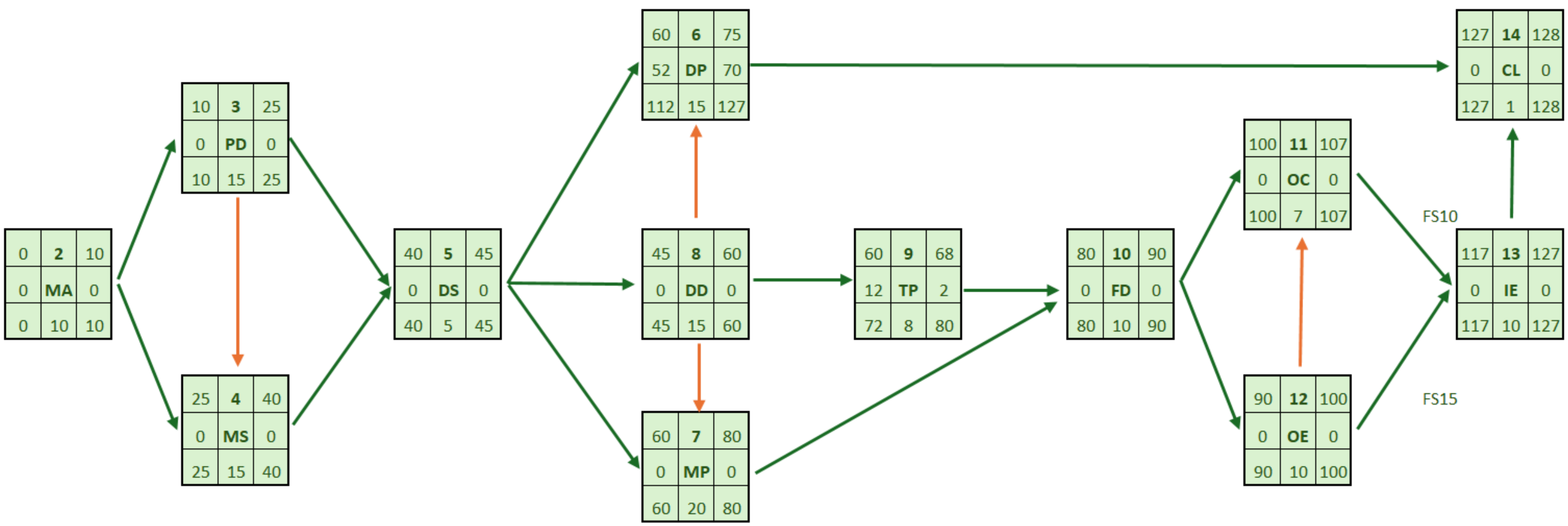


Figure A2: Resource-Leveled Dependencies (After Optimization)

These diagrams visually show how resource leveling adjusted task sequencing to resolve conflicts and shorten the timeline



Note:

Original task dependencies (unchanged)

Adjusted dependencies after resource optimization

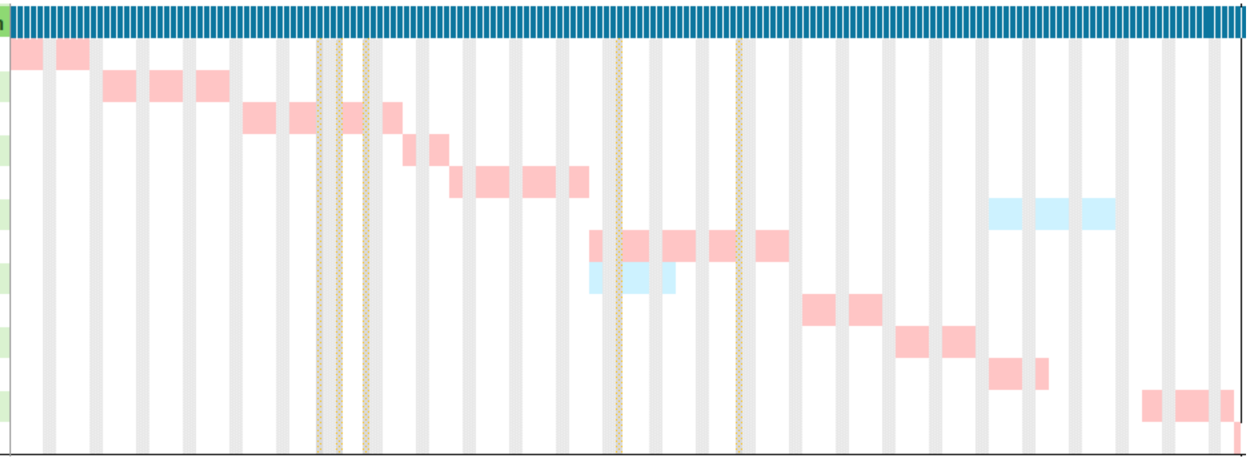
6.2 Appendix B – Detailed Gantt Chart

This Gantt chart illustrates the optimized schedule under Scenario 2, Start and end dates, durations, and dependencies align with the revised sequencing after resource leveling

Highlights:

- *Critical-path activities are marked in pink.*
- *Non-critical activities with slack are shown in blue.*
- *Task 6 (Detailed Marketing Plan) is deliberately delayed by 40 days*

Task_ID	Task Name	Start Date	End Date	Duration
2	Market analysis	3/3/2025	3/14/2025	10
3	Product design	3/17/2025	4/4/2025	15
4	Manufacturing study	4/7/2025	4/30/2025	15
5	Product design selection	5/1/2025	5/7/2025	5
6	Detailed product design	5/8/2025	5/28/2025	15
7	Detailed marketing plan	7/28/2025	8/15/2025	15
8	Manufacturing process	5/29/2025	6/27/2025	20
9	Test prototype	5/29/2025	6/10/2025	8
10	Finalised product design	6/30/2025	7/11/2025	10
11	Order production equipment	7/14/2025	7/25/2025	10
12	Order components	7/28/2025	8/5/2025	7
13	Install production equipment	8/20/2025	9/2/2025	10
14	Celebrate	9/3/2025	9/3/2025	1



6.3 Appendix C – Activities Cost Breakdown

This table summarizes the estimated cost allocation for all project activities under Scenario 2.

Task ID	Task Name	Cost	Percentage
2	Market analysis	$ 20,800	3%
3	Product design	$ 75,600	13%
4	Manufacturing study	$ 56,400	9%
5	Product design selection	$ 28,000	5%
6	Detailed product design	$ 111,000	19%
7	Detailed marketing plan	$ 31,200	5%
8	Manufacturing process	$ 110,400	19%
9	Test prototype	$ 24,320	4%
10	Finalized product design	$ 59,600	10%
11	Order production equipment	$ 3,200	1%
12	Order components	$ 2,240	0%
13	Install production equipment	$ 62,800	11%
14	Celebrate	$ 9,760	2%
	TOTAL	**595,320**	**100%**